Yidnekachew Solomon Haile

Full-Stack & AI Application Engineer

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Full-Stack & AI Application Engineer specializing in building AI-powered web applications that transform raw data into useful, accessible answers. Trained through Evangadi's AI-Powered Full-Stack Application Development program, with hands-on application development across React 19, JavaScript, Node.js, Express 5, and MySQL. Builds AI-enabled application features using the Google Gemini API, text embeddings, semantic search, prompt design, and retrieval-augmented generation (RAG). Combines software development skills with a master's degree in computer science and engineering, a second master's degree in Electronics and Communication Engineering, and enterprise technology experience at the Commercial Bank of Ethiopia. Brings a practical, problem-solving approach to building responsive user interfaces, backend APIs, database-driven applications, and intelligent information retrieval systems.

TECHNICAL SKILLS

- **AI & Machine Learning Integration:** Google Gemini API, Large Language Model (LLM) Integration, Prompt Engineering, Text Embeddings, Semantic Search, Retrieval-Augmented Generation (RAG), Cosine Similarity, Natural Language Processing (NLP), Machine Learning, Deep Learning.
- **Frontend Development:** React 19, JavaScript (ES6+), HTML5, CSS3, Vite, React Router, Responsive Web Design, Component-Based UI Development.
- **Backend & API Development:** Node.js, Express 5, RESTful APIs, Backend Application Logic, JWT Authentication, bcrypt Password Hashing, Client-Server Architecture.
- **Databases & Data Processing**: MySQL, Relational Database Design, SQL, Data Processing, ETL, Apache Spark, Hadoop, Apache Kafka.
- **Tools & Development** Practices: Git, GitHub, API Integration, Software Development Lifecycle (SDLC), Technical Documentation, Requirements Analysis, Agile and Waterfall Methodologies.
- **Programming & Technical Computing:** Python, JavaScript, C++, MATLAB, VHDL, , Plotly, Matplotlib.

PROFESSIONAL TRAINING AND CERTIFICATIONS

AI-Powered Full-Stack Application Development

Evangadi | March 2026–September 2026

- Developed full-stack applications using React 19, JavaScript, Node.js, Express 5, and MySQL, including responsive UI, reusable components, and REST APIs.
- Implemented backend logic, database integration, client-side routing, JWT authentication, and bcrypt password hashing.
- Integrated Google Gemini API, prompt design, text embeddings, semantic search, cosine similarity, and RAG pipelines for AI-powered answers.

KifiyaAI Mastery Training Program

10 Academy – Kifiya AI Mastery | August 2024–November 2024

- **Completed** specialized AI/ML training with distinction, developing knowledge of machine learning methods, analytical problem-solving, and applied AI concepts.

WORK EXPERIENCE

Technical- Information Systems (IS) Officer – Technical Business Analyst

Commercial Bank of Ethiopia (CBE) |, IS-Project Management Offices | IS BA

Addis Ababa, Ethiopia | December 2023–Present

- **Translate** business needs into structured technical requirements and solution specifications, aligning business and IT expectations.

- **Develop and analyze** business cases, RFPs, and technical requirements to support technology evaluation, procurement, and implementation planning.
- **Document and validate** project charters, scope definitions, and technical specifications while coordinating stakeholders to resolve gaps and clarify deliverables.

Part-Time University Lecturer
Addis Ababa University (AAU), Addis Ababa Institute of Technology (AAiT)
Addis Ababa, Ethiopia | December 2022–December 2023

- **Delivered** engineering lectures, practical sessions, and instructional materials for undergraduate students.
- **Guided and mentored** student teams on technical projects, problem-solving, and engineering concepts.

Academic and Research Assistant
Adama Science and Technology University (ASTU)
Adama, Ethiopia | October 2017–2020

- Supported engineering education through laboratory instruction, demonstrations, and manual preparation.
- Analyzed technical data, prepared reports, and maintained laboratory equipment for teaching and research.

Engineering Intern
Dashen Brewery Share Company Ltd.
Debre Birhan, Ethiopia | March 2016–July 2016

- **Collaborated** with control engineers on industrial electrical control systems and automation activities.
- **Gained practical exposure** to PLCs, I/O devices, servo systems, pneumatics, safety circuits, and industrial networks.

EDUCATION

- **MSc**, **Electronics and Communication Engineering** (CGPA= 3.86, with V. Good) September 2020 – July 2022
  - Adama Science and Technology University (ASTU) – Adama, Ethiopia
- **MSc**, **Computer Science and Engineering** (CGPA = 3.72, Good) September 2019 – Mar 2023
  - Adama Science and Technology University (ASTU)– Adama, Ethiopia
- **BSc**, **Electrical and Computer Engineering** (CGPA=3.04) September 2012 – July 2017
  - Addis Ababa University (AAU-AAIT)– Addis Ababa, Ethiopia

ADDITIONAL PROFESSIONAL TRAINING

- **IS Graduate Trainee Program** — Commercial Bank of Ethiopia | December 2023–October 2024
- **Financial Modeling and Valuation Analyst (FMVA®)** — Corporate Finance Institute (CFI), with ALX | September 2022–November 2022

RESEARCH AND PUBLICATIONS

Haile, Y. S., Yetneberk, Z. M., Jadoon, Z., et al. (2025). "Secure Multi-Beam Satellite Systems Using Adaptive Beamforming for 6G Communications." *Proceedings of the 2025 International Conference on Computer, Internet of Things and Smart City (CIoTSC)*, IEEE, pp. 1–5. DOI: https://doi.org/10.1109/CIoTSC67482.2025.11413054

LANGUAGES

- Amharic - Native
- English – Professional Proficiency